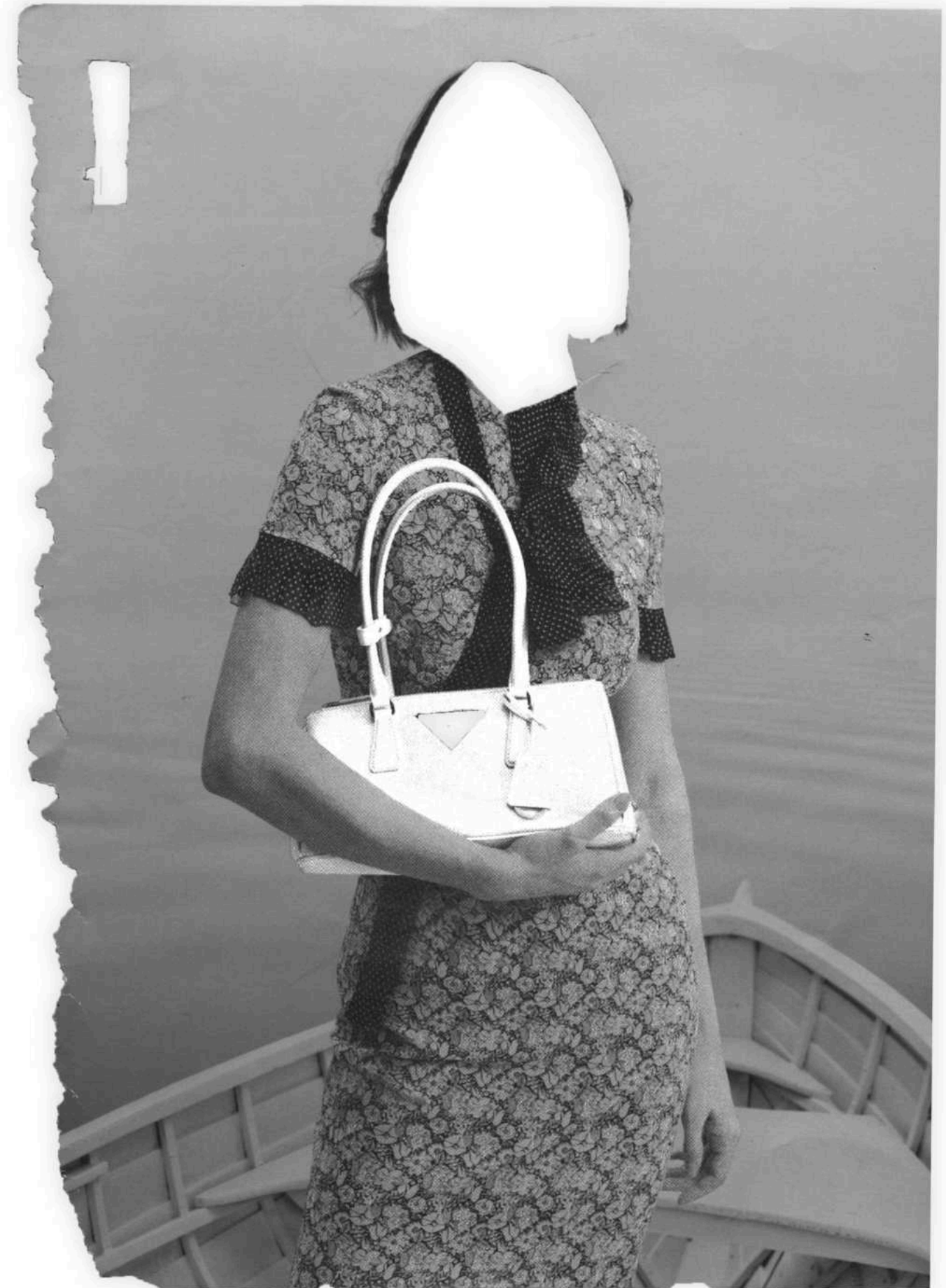
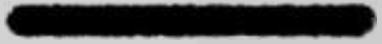


[Pilar Liotta]

From Framing Others to Tracking Perception





Pilar Liotta — MFA Design & Technology student at Parsons

Background: BFA in Graphic Design & Photography, SCAD Atlanta

Focus: Human perception, interaction, and technology

My work shifts between mediums but always explores how design can move, ask questions, and expose invisible systems, aiming toward more intentional and human-centered design.

✦ **Bio**

What is this to you?



SAFE

THREAT

UNKNOWN



✦ 5n5



In my 5-in-5 experiments, I realized I wasn't making five separate projects .
I was asking the same question in five ways.
That led me to merge everything into one concept:

How do we form an opinion about someone when we barely know anything about them?

It's installation art – an interactive perception test disguised as a surveillance system.

The piece uses a webcam and PoseNet tracking to monitor how viewers interact with a series of portraits — images that are either my own photography or scanned, decontextualized magazine cutouts (Harper's Bazaar).

As viewers look, the system tracks their stillness, head movement, and duration of gaze.

At the end, it produces a data-driven dossier that summarizes how they looked — not what they saw.

✦ **Tentative concept**

I looked into how technology changes the way we're seen — and how we see ourselves. We live in a time where being watched feels normal, yet we rarely think about our role in that exchange.

Precedents that informed this work:

Rafael Lozano-Hemmer — Sensor Installations

Turns public participation into data through sensors tracking shadows and heartbeats.

Hans Eijkelboom — People of the Twenty-First Century

Reveals patterns in individuality and social identity through repetition.

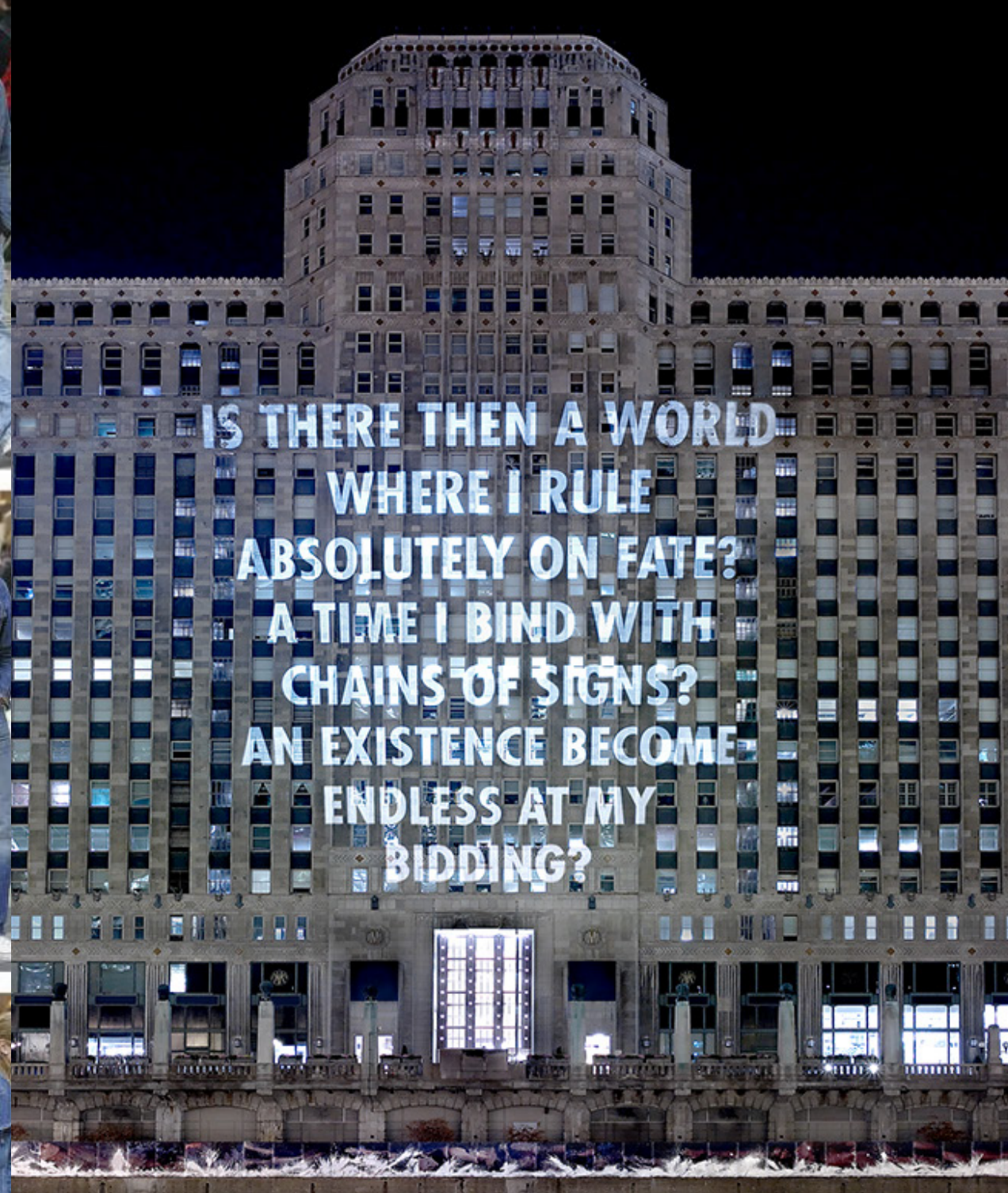
Jenny Holzer — Public Text Projections

Uses language in public space to expose hidden power and provoke reflection.

From these, I began asking:

What happens when the system reflects not the image, but the viewer ; revealing how we perceive, judge, and react?

✦ Precedents & Initial research



✦ Precedents & Initial research

> How do we form an opinion about someone when we barely know anything about them?

> What does it reveal about ourselves when we're aware that we're being watched as we look?

✦ **Design Question**



Qualities:

Ambiguity — letting viewers project their own meanings

Discomfort that leads to reflection

Strategic planning — structured systems that feel controlled yet personal

Humor used as a mirror — softens critique while revealing truth

I want the work to feel:

Minimal — like a stripped-down security interface

Reflective — making viewers aware of their own body and presence

Unsettling but calm — more psychological than dystopian

The aesthetic uses black and white tones, striped visuals, and a quiet surveillance feel, embedding hidden meaning and coded context within familiar systems of observation.

✦ **Design Qualities**

The first version, built in p5.js, tracked how viewers clicked through images — logging time spent and mouse activity to reveal subtle viewing patterns.

In the next stage, I integrated ml5.js PoseNet, shifting from mouse input to webcam tracking.

Now the system:

Tracks the viewer's face and shoulders

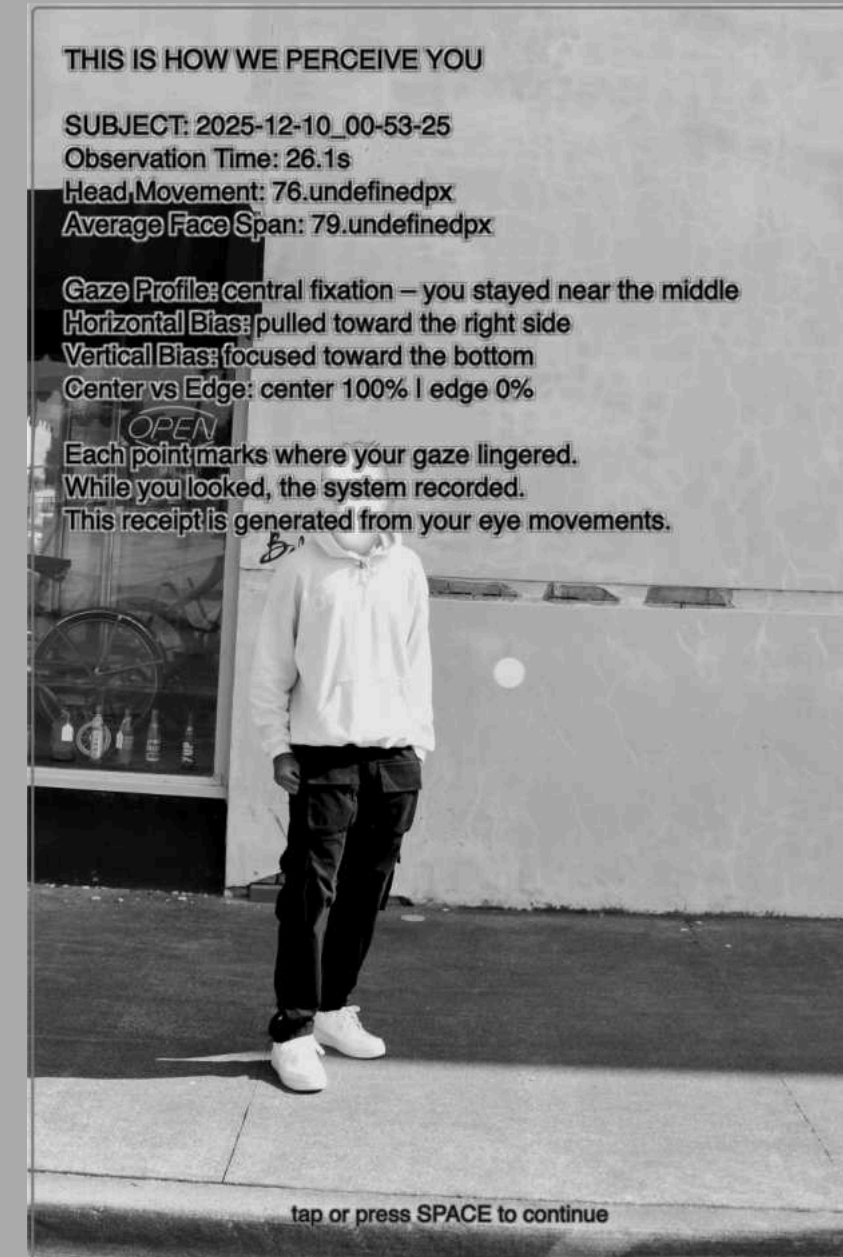
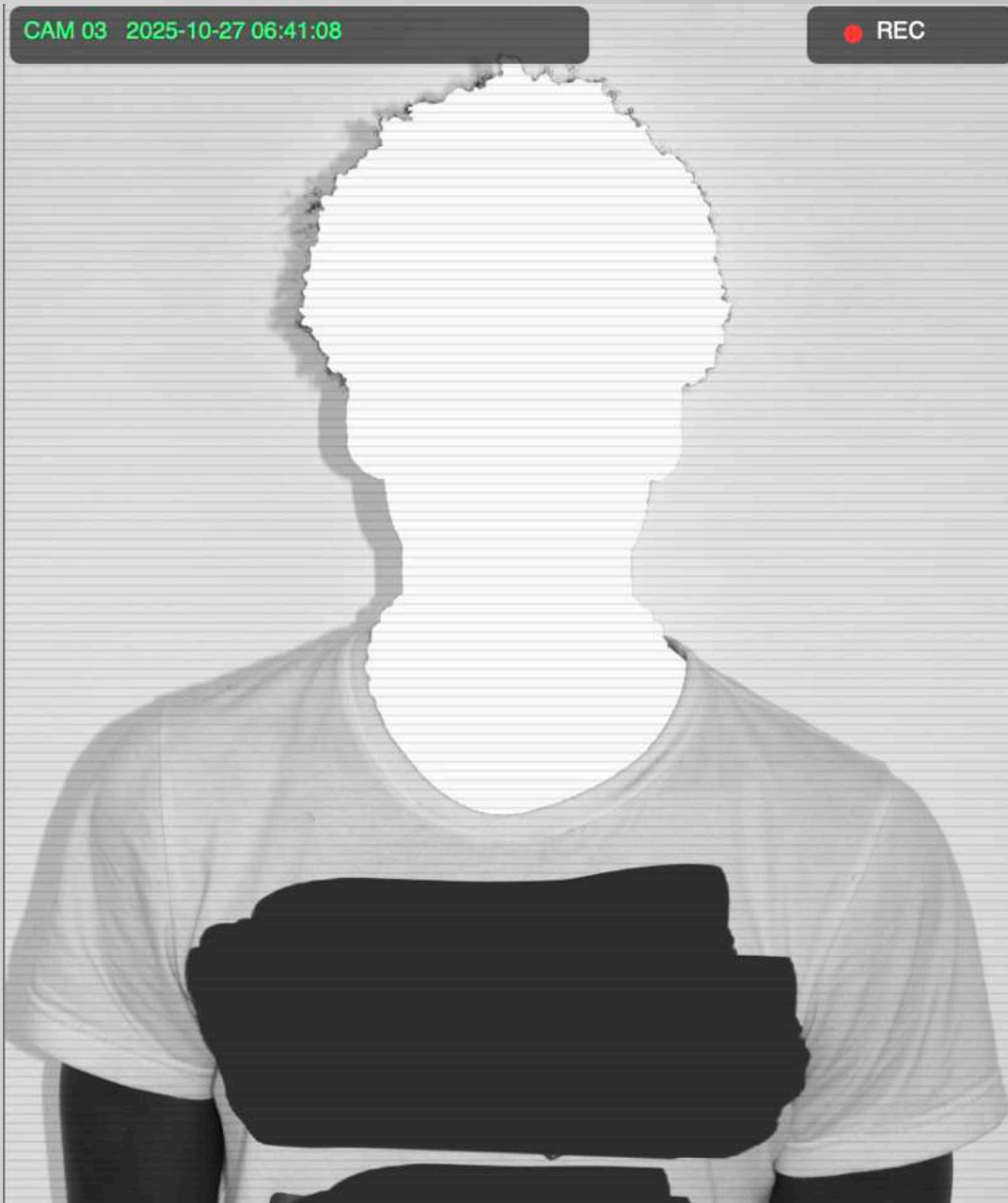
Records time and movement

Displays live messages like "Stillness detected" or "Restless observation"

Ends with a report stamped "YOU HAVE BEEN RECORDED."

It acts as a behavioral mirror; not interpreting emotion, but exposing the act of **perceiving** itself.

✦ Initial Concept Prototype



✦ [Link to P5JS](#)

To simulate the gallery installation I positioned a webcam so it faces the projected image on the screen. The webcam becomes the installation's eye, observing anyone who stands in front of the work. As viewers look at the image, the system tracks their gaze and prints a thermal receipt of their attention, recreating how the piece would function in a real exhibition.

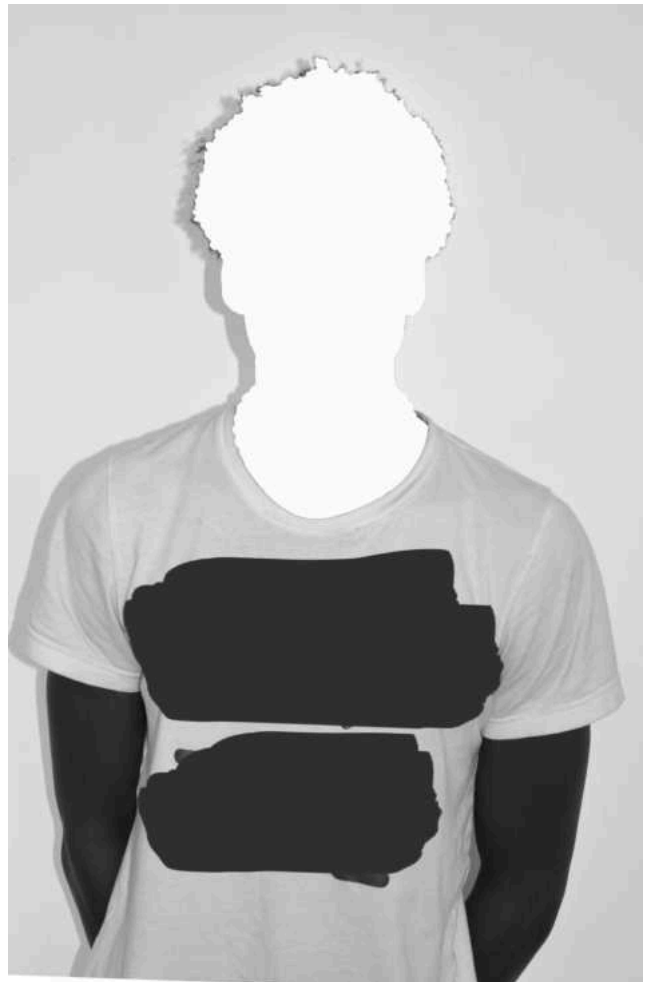
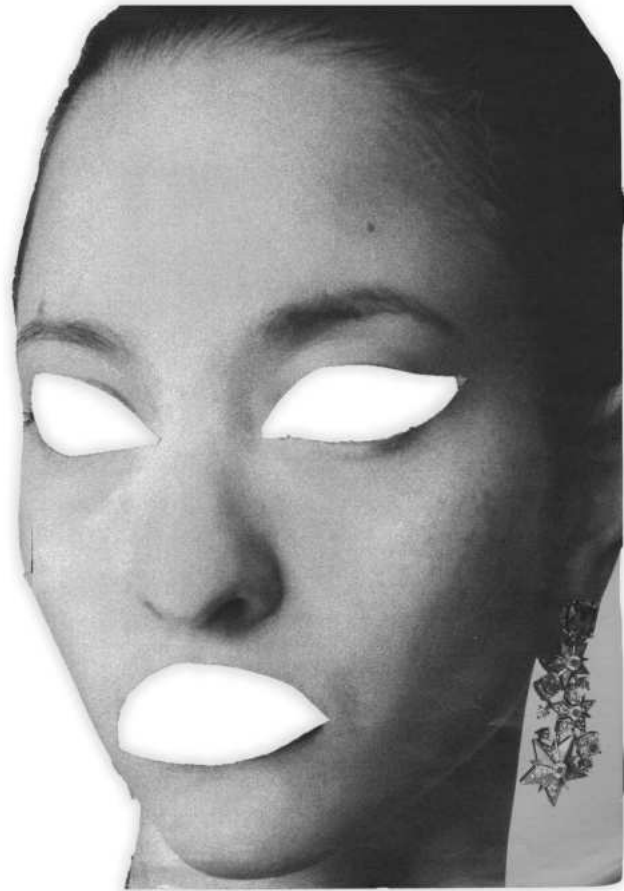
Tools Used –

**p5.js for the interface and receipt drawing
MediaPipe FaceMesh for gaze and blink tracking
GitHub for hosting the prototype
Thermal label printer for the physical receipt
Laptop webcam acting as the gallery camera**

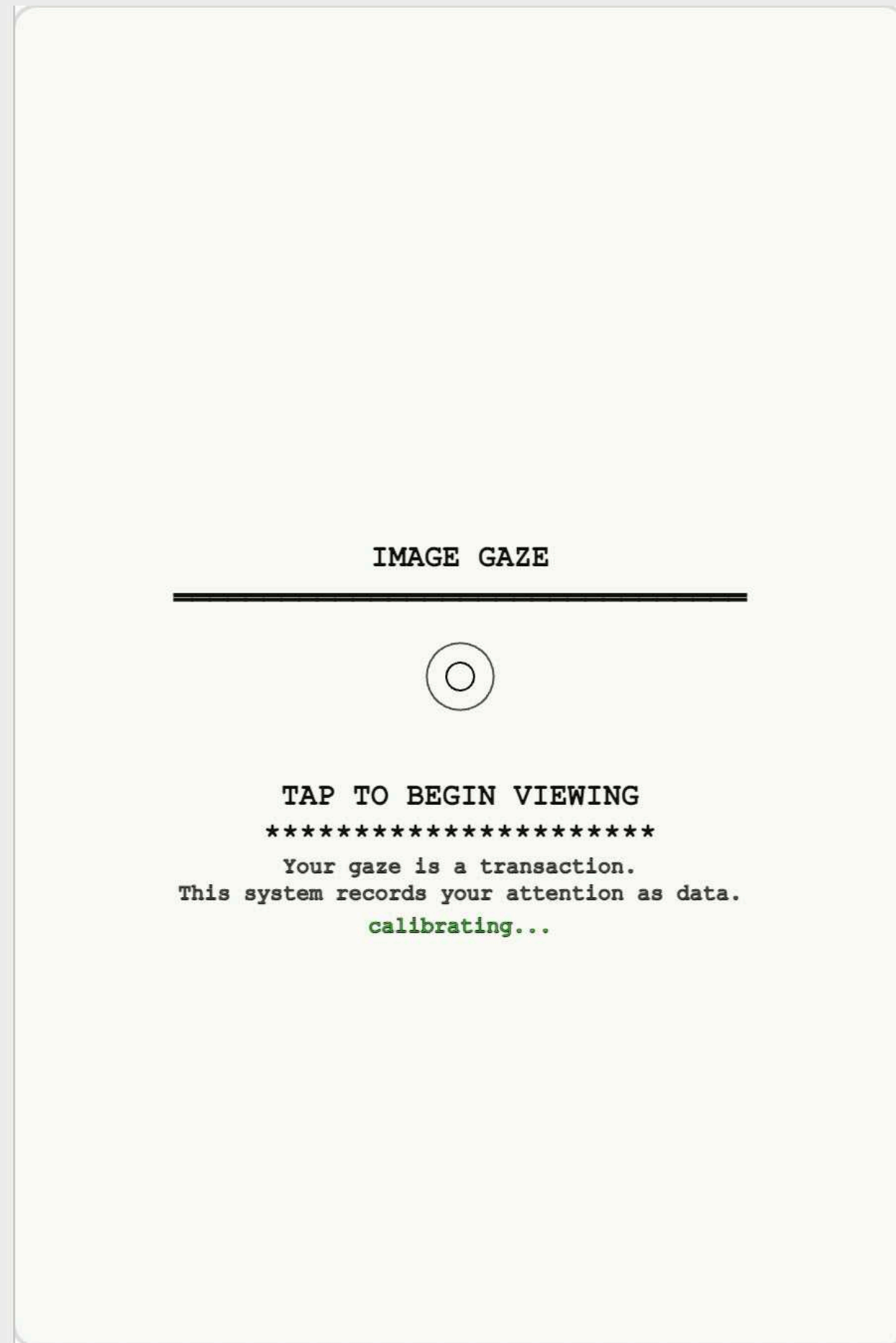
✦ Final Installation



System is recording your gaze



[Link to Project](#)



[Link to Project](#)



LOOKING IS A TRANSACTION. YOUR ATTENTION IS NOW TANGIBLE.

Link to Project

```

*****
IMAGE GAZE RECEIPT
SURVEILLANCE ANALYSIS
*****

DATE: 2025-12-10 01-03-23
SUBJECT ID: -03-Z3
OBSERVATION TIME: 32.8s
FIXATION POINTS: 284
HEAD MOTION: 2400.px
BLINK COUNT: 10
BLINK RATE: 18.2/min

BLINK BEHAVIOR:
Behavior: NORMAL

GAZE ANALYSIS:
Profile: CENTRAL FIXATOR
H-Bias: Left-biased
V-Bias: Upper-focused
Center: 94% | Edge: 1%

GAZE PATTERN MAP:

```

```

O = Blink detected - Each Dot marks fixation point
-----
Data has been captured and stored.

YOUR GAZE IS A TRANSACTION.

THIS RECEIPT IS A RECORD OF YOUR ATTENTION.

LOOKING IS A TRANSACTION. YOUR ATTENTION IS NOW TANGIBLE.
-----

```

This installation reflects how we perceive and are perceived. We form instant interpretations of art and people, and the system mirrors that back by watching the viewer. Your gaze becomes data, and the printed receipt shows how easily attention turns into a transaction.

✦ **Final**

✦ **Questions?**